Calibrating a pandemic optimal control model: empirical parameter estimates

A discrete-time quarterly LQR/iLQR social planner model for COVID-19 requires 13 parameters spanning macroeconomic dynamics, epidemiology, fiscal policy, and public finance. This report provides empirical estimates from the academic literature for each parameter, calibrated for OECD economies during 2020–2021. Where direct estimates exist, specific values with sources and methodology are given; where parameters must be inferred indirectly, calibration strategies are proposed. A summary calibration table and a comprehensive COVID-19 wave timeline appear at the end.

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1. Output gap persistence and lockdown-induced hysteresis

ρ_y — Baseline quarterly AR(1) coefficient of the output gap

This parameter governs how quickly output gaps close absent lockdown-specific scarring. The literature provides a well-established range.

Smets & Wouters (2007), the workhorse Bayesian DSGE model estimated on US quarterly data 1966–2004, (American Economic Association) yields habit persistence **h ≈ 0.71** and technology shock persistence **ρ_a ≈ 0.95**. The reduced-form output gap autocorrelation implied by this model is approximately **0.88–0.92**. Galí's (2015) textbook New Keynesian model (Google Books) with habit formation h = 0.7 and demand shock persistence ρ = 0.9 produces quarterly output gap autocorrelations of **0.90–0.95**. Reduced-form VAR estimates of US output gap persistence cluster around **0.85–0.95** quarterly.

At the higher end, **Cerra & Saxena (2008)** find that output losses from financial crises are essentially permanent, (American Economic Association) implying near-unit-root persistence (ρ ≈ 0.95–0.997). Their updated survey with Fatás (Cerra, Fatás & Saxena, 2023, JEL) confirms that 60–80% of output movements at the 2–4 year horizon reflect permanent components. (IMF) **Kozlowski, Veldkamp & Venkateswaran (2020)** find that belief-scarring from COVID implies GDP remains ~4% below the pre-pandemic steady state permanently, (IDEAS/RePEc) (nber) consistent with ρ ≈ 0.98–0.99 when belief effects are included. The **IMF WEO April 2021** projects advanced-economy scarring of ~2% of GDP, (International Monetary Fund) implying a blended persistence of **0.93–0.97**. (imf)

However, ρ_y in this model should capture *baseline cyclical persistence before lockdown scarring* (which is handled by ψ). The appropriate calibration therefore draws from DSGE and VAR estimates rather than post-crisis hysteresis studies.

Source	ρ_y range	Context
Standard NK, no habit, moderate shocks	0.85–0.90	Pure cyclical
Smets-Wouters (2007)	0.88–0.92	Bayesian DSGE
Galí (2015) with habit h = 0.7	0.90–0.95	Extended NK
Reduced-form VARs	0.85–0.95	Empirical
IMF WEO-implied (with scarring)	0.93–0.97	Post-crisis

Recommended calibration: ρ_y = 0.90 (range 0.85–0.95). This implies a half-life of **~6.6 quarters** (~1.7 years), consistent with mainstream DSGE calibrations while excluding lockdown-specific scarring.

ψ — Lockdown-induced hysteresis sensitivity

No paper directly estimates ψ in the specified formulation (effective persistence = ρ_y + ψ·S_t). All estimates below are translations from related empirical findings.

Barrero, Bloom & Davis (2021) find that **32–42% of COVID-induced layoffs became permanent**, NBER Becker Friedman Institute compared to ~15% in normal recessions. The additional 27 percentage points of permanent job destruction represent lockdown-specific scarring. During peak containment (S ≈ 0.8), approximately 1.3% of potential output was permanently destroyed per quarter of full lockdown, translating to **ψ ≈ 0.03–0.05**. **Guerrieri, Lorenzoni, Straub & Werning (2022, AER)** establish that Keynesian supply shocks from sectoral shutdowns amplify output losses beyond the direct supply effect, American Economic Association supporting values at the higher end of the range.

Chetty et al. (2020) document that ~19% of small businesses closed by mid-2020, Evergreensmallbusiness with employment falling 37% for bottom-quartile workers. NBER However, they find much of the economic impact was *voluntary* (fear-driven), not solely from mandated lockdowns — an important caveat that pushes ψ downward. Federal Reserve establishment data show ~300,000 permanent business closures in Q2 2020 (destroying ~1.2M jobs), Federal Reserve suggesting **ψ ≈ 0.01–0.03** from the business exit channel alone.

Cross-country evidence from the **IMF WEO April 2021** showing ~2% advanced-economy scarring, imf International Monetary Fund with average containment S ≈ 0.4 over ~6 quarters, implies **ψ ≈ 0.02–0.06**. Learning-by-doing models (Engler & Tervala, 2022) with TFP scarring ratios of 0.42 suggest higher values of **ψ ≈ 0.05–0.10**. PubMed Central

Recommended calibration: ψ = 0.05 (range 0.02–0.10). At full lockdown (S = 1), effective persistence rises to 0.95, roughly doubling the output gap half-life from 6.6 to 13.5 quarters. **Calibration strategy**: set ρ_y = 0.90 from DSGE literature, then calibrate ψ so that model-predicted scarring given observed containment paths matches IMF medium-term output loss estimates of ~2–3% for advanced economies. Sensitivity analysis over ψ ∈ [0.02, 0.10] is advisable.

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2. Epidemiological parameters: reproduction and fatality

ρ_θ — Quarterly growth rate of infection prevalence (uncontrolled)

The model's infection dynamics follow $\theta_{t+1} = \rho_\theta \cdot \theta_t \cdot (1 - \varphi_S \cdot S_t)$. The parameter ρ_θ captures uncontrolled quarterly infection growth.

Consensus R_0 estimates by variant are well established. For the **ancestral strain**, the WHO consensus and meta-analyses (Li et al. 2020, NEJM; Alimohamadi et al. 2020, PLoS ONE) place R_0 at **2.5–3.0**. **Davies et al. (2021, Science)** found **Alpha** was 43–90% more transmissible, implying $R_0 \approx$ **4.0–4.5**. (Science) (PubMed) **Liu & Rocklöv (2021, J Travel Med)** estimated **Delta** R_0 = **5.08**. (PubMed) **Liu et al. (2022)** meta-analyzed **Omicron** at $R_0 \approx$ **9.5** (range 5.5–24).

Converting R_0 to quarterly growth is where complications arise. In an SIR framework with generation time ~5 days, a quarter contains ~18 generations, giving uncontrolled growth factors of R_0^18 — which for $R_0 = 2.5$ yields $\sim 3.8 \times 10^7$. Alternatively, with infectious period $D \approx 10$ days and daily growth rate $r = (R_0-1)/D$, the quarterly growth factor is $\exp(90 \cdot r)$, producing similarly astronomical numbers. **These enormous factors confirm that containment ($\varphi_S \cdot S_t$) must be very close to 1 for the linearized model to produce realistic dynamics.**

Variant	R_0	Quarterly growth factor (uncontrolled)
Original	2.5–3.0	$\sim 10^7$–10^8 (generation method)
Alpha	4.0–4.5	$\sim 10^{10}$–10^{11}
Delta	5.0	$\sim 10^{12}$–10^{15}
Omicron	8–10	$\sim 10^{16}$–10^{33}

Calibration strategy: ρ_θ cannot be taken literally from SIR exponential growth over 90 days, as susceptible depletion intervenes. For the reduced-form quarterly model, two approaches work: (1) calibrate $\rho_\theta \approx$ **2–10** as an effective quarterly reproduction factor, with $\varphi_S \cdot S$ absorbing most growth dampening; (2) use a log-linearized version where ρ_θ represents the *net* quarterly growth rate in a partially controlled environment, calibrated to match observed wave dynamics. The model-builder should choose ρ_θ and φ_S jointly to replicate observed quarterly infection trajectories. A practical starting point: **$\rho_\theta \approx$ 3–5** for the ancestral/Alpha strain, scaling upward for Delta and Omicron.

δ_θ — Infection fatality rate (IFR)

The pre-vaccination OECD IFR is well characterized. **Meyerowitz-Katz & Merone (2020)** meta-analyzed 24 seroprevalence studies and found an overall IFR of **0.68%** (95% CI: 0.53–0.82%). **Levin et al. (2020, Eur J Epidemiol)** — restricted to OECD countries — found an exponential age-IFR relationship where 90% of cross-country variation was explained by age structure alone. For a typical OECD country with ~19% of population aged 65+, the implied population IFR is **0.8–1.15%**. **Brazeau et al. (2020, Imperial College)** estimated **1.15%** for high-income country age structures. The **COVID-19 Forecasting Team (Lancet, 2022)** documented a

~33% IFR decline from April 2020 (**0.47%** globally) to January 2021 (**0.31%**), driven by improved treatment (dexamethasone). The Lancet **Msemburi et al. (2023, Nature)** estimated **14.83 million** global excess deaths in 2020–2021, 2.74× reported COVID deaths. ResearchGate

IFR declined dramatically across waves due to vaccination, prior immunity, and improved treatment. Italian longitudinal data (Marziano et al., 2023) provides the clearest trajectory: PubMed Central

Wave/variant	IFR	Change from first wave
Ancestral, first wave (Q1–Q2 2020)	2.2% (Italy)	Baseline
Ancestral, second wave (Q3–Q4 2020)	1.0%	−54%
Alpha (Q1–Q2 2021)	0.39%	−82%
Delta (Q2–Q3 2021)	0.12%	−95%
Omicron (Q4 2021+)	0.02%	−99%

Recommended calibration: δ_θ = 0.008–0.010 (0.8–1.0%) for the pre-vaccination OECD baseline. For time-varying calibration: 0.8–1.0% (Q1–Q2 2020), 0.5–0.8% (Q3–Q4 2020), 0.3–0.5% (Q1 2021, Alpha + early vaccination), 0.1–0.3% (Q2–Q3 2021, Delta + significant vaccination), 0.02–0.07% (Q4 2021+, Omicron).

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3. NPI effects on output and transmission

α_S — Contractionary effect of full lockdown on quarterly GDP

The key empirical challenge is decomposing total pandemic economic losses into lockdown-mandated versus voluntary behavioral responses.

Deb, Furceri, Ostry & Toscani (2021, IMF) estimate that containment measures caused ~**10–15% loss in industrial production** over 30 days. SSRN In countries with limited fiscal stimulus, the effect reached **~29%**. PubMed Central **Coibion, Gorodnichenko & Weber (2020)** find aggregate consumer spending dropped **31 log points** between January and April 2020, implying a ~**10–14% quarterly GDP decline** if extrapolated. **Chetty et al. (2020)** tracked a ~30% peak monthly spending decline using real-time data, NBER consistent with ~10–15% quarterly GDP loss.

However, **Goolsbee & Syverson (2021, J Pub Econ)** decompose the decline: total foot traffic fell 60 percentage points, but shelter-in-place orders accounted for only **7 percentage points** (~12% of the total). SSRN The remaining 88% was voluntary. **Born, Dietrich & Müller (2021)** use Sweden as a natural experiment and find that a 9-week lockdown would have reduced GDP by only **~1.6 percentage points** in Q1 and **~0.7 pp** in Q2 beyond what Sweden experienced without one. PLOS **Eichenbaum, Rebelo & Trabandt (2021)** model the incremental GDP cost of optimal containment Oxford Academic at ~**8–15 pp** beyond voluntary behavioral effects. Oxford Academic NBER

Source	Quarterly GDP loss (full lockdown)	Notes
Deb et al. (2021, IMF)	10–15%	Industrial production proxy
Coibion et al. (2020)	10–14%	Survey-based spending
Goolsbee & Syverson (2021)	1–2% (marginal)	Policy effect only; 88% voluntary
Born et al. (2021)	1–2% (marginal)	Sweden counterfactual
Eichenbaum et al. (2021)	8–15% (incremental)	SIR-macro model

Recommended calibration: α_S = 12 (range 8–15). This assumes S captures the full containment regime including induced behavioral changes. If S represents only mandated policy (separating voluntary responses modeled elsewhere), use α_S ≈ 3–5. The Oxford Stringency Index maps approximately as S = OxCGRT/100, with "full lockdown" corresponding to scores of 80–90 (i.e., S ≈ 0.85–0.95 in practice).

φ_S — NPI effectiveness in reducing R_t

Flaxman et al. (2020, Nature) found lockdowns reduced R_t by **81%** (75–87%) across 11 European countries, PubMed though this estimate faced criticism for attributing the entire observed R_t decline to the last-implemented NPI. **Brauner et al. (2021, Science)** — the most comprehensive study (41 countries) — estimated individual NPI effects that combine multiplicatively Science to approximately **77–82% total reduction**: gathering limits ≤10 (−36%), school closures (−38%), business closures (−40%), stay-at-home orders (−13%). Science Science **Haug et al. (2020, Nature Human Behaviour)** found more conservative estimates using four computational methods, Nature ResearchGate with combined effects of approximately **30–60%** reduction. The UK empirical experience confirmed R_t declining from ~2.7 to ~0.6 during lockdown, a **~78% reduction**.

Sharma et al. (2021, Nature Communications) found NPI effectiveness was **lower in subsequent waves** due to behavioral adaptation, Nature implying φ_S is time-varying and declining.

Recommended calibration: φ_S = 0.70 (range 0.60–0.82). With R_0 = 3.0 and φ_S = 0.70, full lockdown yields R_t = 3.0 × 0.30 = 0.90, consistent with observed sub-unity R_t values during spring 2020 lockdowns.

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4. Fiscal multipliers and the cost of demand injection

α_F_DI — Fiscal multiplier for direct transfers

Pre-pandemic estimates bracket the normal-times transfer multiplier. **Ramey & Zubairy (2018, JPE)** find multipliers **below unity** (0.6–0.8) regardless of slack; IDEAS/RePEc **Auerbach & Gorodnichenko (2012)** find recession multipliers of **1.0–1.5**; CEPR **Christiano, Eichenbaum & Rebelo (2011)** derive ZLB multipliers of **2–5** for government purchases Wiley Online Library (lower for transfers, attenuated by MPC).

Pandemic-specific evidence reveals substantially lower multipliers for transfers. **Coibion, Gorodnichenko & Weber (2020)** find only **15%** of recipients mostly spent their stimulus checks; the aggregate MPC was

approximately **0.40**. (CEPR) (Chicago Booth Review) **Parker, Schild, Erhard & Johnson (2022, BPEA)** track EIP spending: Round 1 had an MPC of ~10% on nondurables within 3 months, (SSRN) rising to **~45% cumulatively over 6 months**. Round 2 showed ~60% cumulative spending. Round 3 (Spring 2021) showed **essentially zero spending response**, as household liquidity was already ample. (brookings) **Baker et al. (2020)** found spending of **$0.25–$0.40 per dollar** in the first weeks, concentrated among liquidity-constrained households. (NBER)

The **CBO** estimated COVID transfer multipliers at **0.3–0.6**, noting they were "partially tempered by social distancing." (Committee for a Responsible F…) **Auerbach et al. (2022)** found that lockdowns completely shut down the fiscal transmission mechanism — "no detectable employment response in cities under stay-at-home orders." (CEPR +2) The **IMF (WP 2022/149)** finds cumulative multipliers one year after a health crisis are **about twice larger** than contemporaneous multipliers, as pent-up demand is released upon reopening.

Recommended calibration: α_F_DI = 0.5 for impact (same quarter), **0.7** cumulative over 4 quarters (range 0.3–1.0). Pandemic transfer multipliers were substantially below normal-recession values due to lockdown constraints, broad targeting, and precautionary saving.

κ_F_DI — Budgetary cost of demand injection

US Economic Impact Payments were structured as **non-taxable advance refundable tax credits** — no income tax clawback, no repayment obligation. (Slvfed) The gross fiscal cost is therefore 1.0 per dollar disbursed.

The net fiscal cost accounts for revenue feedback: **κ_F_DI = 1 − τ × m**, where τ is the effective tax rate and m the multiplier. With OECD-average τ ≈ 0.34 (OECD) and pandemic multiplier m ≈ 0.5–0.7, revenue feedback is 0.17–0.24, giving **κ_F_DI ≈ 0.76–0.83**. For the US (τ ≈ 0.27): κ ≈ 0.81–0.87. The **CBO** cross-check: $2.6T COVID relief boosted GDP by $1.5T (multiplier ~0.6), (Committee for a Responsible F…) with US revenue feedback of 0.6 × 0.27 = 0.16, yielding κ = 0.84.

Recommended calibration: κ_F_DI = 0.80 (OECD average; range 0.70–0.90). During active lockdowns (low multiplier), κ approaches 0.90; as the economy reopens, κ falls toward 0.70–0.75.

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5. Capacity preservation: effectiveness and fiscal cost

η — Effectiveness of job retention schemes in reducing scarring

Capacity preservation instruments (Kurzarbeit, furlough, PPP) reduce output persistence by keeping firms alive and workers attached to jobs during lockdowns.

On short-time work, **Cahuc, Kramarz & Nevoux (2021)** find French STW saved approximately **1 in 6** participating jobs, with higher firm survival rates. **Kopp & Siegenthaler (2021)** estimate Swiss STW saved **1 in 3** jobs during the Great Recession. (PubMed Central) **Bennedsen et al. (2020)** find **1 in 3** jobs saved at supported Danish firms. (PubMed Central) The **OECD (2020)** reports job retention schemes covered ~**50 million jobs** (20% of dependent employment) at peak. **Ando et al. (2022, IMF)** estimate unemployment would have been **2.5 percentage points higher** across the euro area without these schemes. (International Monetary Fund)

For the US approach, **Autor et al. (2022, JEP)** find PPP preserved **2–3 million job-years** at a cost of **$169K–$258K per job-year**, with only 23–34% of PPP dollars reaching workers who would otherwise have lost jobs.

American Economic Association **Granja et al. (2022)** found modest employment effects (about 2% boost at small businesses) relative to the program's $525B scale. American Experiment

Comparative evidence is striking: COVID employment recovered in **29 months** from a 14.7% US unemployment trough, versus **76 months** after the GFC's shallower 6.3% trough. Socialcatalystlab European countries with strong STW saw barely any unemployment rise (adjusted for hours), suggesting capacity preservation offset 60–80% of potential scarring. European Central Bank

Recommended calibration: η = 0.05 (range 0.03–0.08). This implies maximum capacity preservation (F_CP = 1) offsets the scarring effect of ψ = 0.05 at full lockdown, consistent with the European experience. **Calibration strategy**: match the differential scarring outcomes between countries with strong capacity preservation (Europe) versus those relying more on demand injection (US), adjusting for underlying shock severity.

κ_F_CP — Budgetary cost of capacity preservation

The fiscal cost varies enormously by instrument type. **Short-time work subsidies** are pure fiscal outflows (**κ = 1.0**); Germany spent ~€30B+ on Kurzarbeit, ProPublica and the UK's CJRS cost ~£61B. **US PPP loans** had a **96% forgiveness rate** (SBA data, October 2022), making them effectively grants (**κ ≈ 0.96**). NPR St. Louis Fed

European loan guarantee schemes were far cheaper. **UK BBLS** (100% government guarantee, £46.5B drawn): IBISWorld OBR lifetime write-off estimate of **42–46%** of drawn value. UK Parliament GOV.UK **UK CBILS/CLBILS** (80% guarantee, £30B drawn): default rate only ~**3–7%**. **France PGE** (70–90% guarantee, €144B disbursed): FranceTransactions guarantee calls below **4%** of total as of end-2024, with ~76% already repaid. French Banking Federation German KfW guarantees showed estimated losses of **5–15%**.

Instrument	κ (net cost per unit deployed)
Short-time work / wage subsidies	1.00
US PPP (forgivable loans)	0.96
UK BBLS (100% guarantee)	0.42–0.46
UK CBILS (80% guarantee)	0.03–0.07
France PGE (70–90% guarantee)	0.04–0.08
European loan guarantees (typical)	0.05–0.20

Using approximate OECD fiscal-weight shares (45% STW/wage subsidies, 15% forgivable loans, 30% loan guarantees, 10% direct loans): **weighted average κ_F_CP ≈ 0.66**. European-only portfolios yield ~0.61; US-weighted portfolios ~0.89.

Recommended calibration: κ_F_CP = 0.60 (range 0.50–0.70 for OECD average). This reflects that while loan guarantees had enormous headline envelopes, their actual fiscal cost was remarkably low.

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6. Sovereign debt dynamics, automatic stabilizers, and health costs

r — Real interest rate on sovereign debt

During 2020–2021, real sovereign borrowing rates were **deeply negative** across the OECD due to central bank QE, flight to safety, and ZLB forward guidance. US 10-year TIPS real yields averaged approximately **−1.0%** in 2020 and **−0.8% to −1.1%** in 2021. German real yields were approximately **−1.8% to −2.0%**. The GDP-weighted OECD average real rate on new issuance was approximately **−1.0% to −1.2%** annually.

Rachel & Summers (2019, BPEA) estimate the advanced-economy neutral real rate declined by at least **300 basis points** since the 1970s, reaching approximately **0%** by the late 2010s. NBER **Blanchard (2019, AER)** documents that r < g has been "more the historical norm than the exception" since 1950. American Economic Association Paulgp **Jordà et al. (2019, QJE)** show long-run real government bond returns of 1–3% historically, but near zero by the 2010s.

The effective (average) real rate on outstanding debt was higher than the marginal rate due to legacy bonds, approximately **0.5–0.8%** nominal effective rate minus inflation, yielding roughly **−0.3% to +0.5%** in 2020 and **−1.0% to −1.5%** in 2021 (as inflation eroded fixed coupons). The r − g differential averaged approximately **−1.5% to −2.0%** over 2020–2021, highly favorable for debt sustainability.

Recommended calibration: r = −0.25% per quarter (i.e., −1.0% annually; range −0.13% to −0.38% quarterly). This uses the marginal new-issuance real rate. If calibrating to the effective rate on the debt stock, use r ≈ 0.0% to +0.1% per quarter.

γ_y — Budget semi-elasticity

The budget semi-elasticity measures how much the deficit widens (in pp of GDP) per 1 pp increase in the output gap. **Girouard & André (2005, OECD)** established the foundational methodology. **Price, Dang & Botev (2015, OECD WP 1275)** provided updated estimates. OECD **Mourre et al. (2019, European Commission)** recalibrated for EU member states.

Country-specific values vary substantially with the size of the welfare state: **Denmark ~0.75–0.80**, **Sweden/Belgium ~0.60–0.65**, **France/Netherlands ~0.55–0.60**, **Germany/Italy ~0.50–0.55**, **UK/Spain ~0.45–0.50**, **US ~0.34–0.37**, **Japan ~0.35–0.40**. The **unweighted OECD average** is approximately **0.44–0.48**; IMF the **GDP-weighted average** is slightly lower at **0.42–0.46** due to the US's large weight and below-average semi-elasticity; the **euro area average** is approximately **0.52–0.54**.

The ECB (Economic Bulletin 6/2020) confirmed these estimates in practice: automatic stabilizers accounted for ~2.8% of GDP of the 8.5% euro area deficit in 2020, European Central Bank consistent with a semi-elasticity of ~0.52 and an output gap of ~−5%. **No quarterly conversion is needed** — the semi-elasticity is a ratio, not a flow.

Recommended calibration: γ_y = 0.50 (range 0.42–0.55 for OECD average; US-specific: 0.35; euro area: 0.53).

c_H — Pandemic health expenditure per unit of infection prevalence

OECD health spending grew **5.0% in real terms** in 2020 OECD while GDP fell ~4.5%, pushing the health-to-

GDP ratio up by approximately **1 percentage point** (OECD) (from 8.8% to ~9.7%). COVID-specific spending represented an estimated **6% of public health spending in 2020** and nearly **10% in 2021** (Healthsystemsfacts) (OECD Health Statistics 2024), translating to approximately **0.35–0.50% of GDP** in 2020 and **0.55–0.80% of GDP** in 2021.

Bottom-up cost estimates yield per-infection costs of approximately **$1,700** in OECD countries: hospitalization (7.5% rate × $12,500 average = $937), ICU (1.5% × $30,000 = $450), outpatient care (~$200), and testing (~$150). US per-patient inpatient costs averaged **$24,826** (Ohsfeldt et al., 2021), (PubMed Central) roughly 2× the OECD average after PPP adjustment.

To calibrate c_H where extra health spending (% of GDP) = c_H × θ (quarterly average point prevalence as share of population): using top-down OECD data, quarterly COVID health spending of ~0.10–0.175% of GDP with average point prevalence of ~0.5–1.5% yields **c_H ≈ 10–20**. Bottom-up per-infection costs plus overhead (vaccination, PPE, public health infrastructure) give **c_H ≈ 8.5–12.7**.

Recommended calibration: c_H = 10 (range 5–20, expressed as percentage points of GDP per unit of prevalence per quarter). This is sensitive to the definition of θ: if θ represents quarterly *new* infections rather than point prevalence (which is ~3–5× higher due to ~10-day infectious duration in a 90-day quarter), use c_H ≈ 2–4 instead.

COVID-19 pandemic wave timeline for OECD countries

The pandemic unfolded across four major waves, each driven by a distinct variant with sharply different epidemiological characteristics.

Wave 1 (Q1–Q2 2020, original strain) peaked in March–April across most OECD countries, with estimated weekly prevalence of **0.2–1.0%** of population in severely affected areas (Lombardy, New York). Cumulative attack rates by end of Wave 1 were 1–5% in most OECD countries. The IFR was **0.8–1.15%** (higher where healthcare was overwhelmed, e.g., Italy first wave ~2.2%). (Nature) Infection hospitalization ratio was approximately **3–5%**. No vaccines were available. Severe lockdowns (Oxford Stringency Index 80–90+) were implemented across virtually all OECD countries from March–May.

Wave 2 (Wikipedia) **(Q4 2020–Q1 2021, Alpha emergence)** was driven first by the original strain, then by **Alpha (B.1.1.7)**, which emerged in the UK in September 2020 and became dominant across Europe and the US by March 2021. Weekly case rates reached **500–1,500 per million** in many European countries; the UK REACT-1 study showed peak prevalence of ~1.5–2%. IFR had declined to **0.4–0.7%** due to improved treatment (dexamethasone, introduced summer 2020). Vaccination began December 8, 2020 (UK first); by end of Q1 2021, OECD average first-dose coverage was **~10–15%** (Israel ~55%, UK ~45%, US ~28%).

Wave 3 (Q2–Q3 2021, Delta) was dominated by **Delta (B.1.617.2)**, which became globally dominant by June–July 2021. Weekly prevalence reached **0.3–1.0%** in severely hit OECD countries. Effective IFR dropped to **0.1–0.3%** as vaccination coverage climbed to **40–70%** first-dose by mid-2021 and **55–60%** fully vaccinated by September. Delta was intrinsically ~40–60% more severe than Alpha but vaccines were 93% effective against hospitalization. This wave hit the unvaccinated disproportionately.

Wave 4 (Q4 2021–Q1 2022, Omicron) produced by far the largest infection surge. **Omicron (B.1.1.529)** drove peak weekly prevalence to **3–7%** in many OECD countries (UK ONS: ~1 in 15 in England, January 2022). Despite record case counts, IFR fell to **0.02–0.07%** due to lower intrinsic severity, high vaccination (70–80% fully vaccinated, 20–55% boosted), and prior immunity. Hospitalization rates dropped to ~0.2–0.5% of infections. Many countries shifted from containment to "living with COVID" during this wave.

Wave	Timing	Variant	R_0	Peak weekly prevalence	IFR	Vaccination (1+ dose)
1	Q1–Q2 2020	Original	2.5–3.0	0.2–1.0%	0.8–1.15%	0%
2	Q4 2020–Q1 2021	Original→Alpha	3.5–4.5	0.5–2.0%	0.4–0.7%	0–15%
3	Q2–Q3 2021	Delta	5.0	0.3–1.0%	0.1–0.3%	40–70%
4	Q4 2021–Q1 2022	Omicron	8–10	3–7%	0.02–0.07%	70–80%

Master calibration table

Parameter	Symbol	Central value	Range	Key sources
Output gap persistence	ρ_y	**0.90**	0.85–0.95	Smets & Wouters (2007); Galí (2015); reduced-form VARs
Lockdown hysteresis	ψ	**0.05**	0.02–0.10	Barrero et al. (2021); Chetty et al. (2020); IMF WEO 2021
Quarterly infection growth	ρ_θ	**3–5** (effective)	Model-dependent	Li et al. (2020); Davies et al. (2021); Liu & Rocklöv (2021)
Infection fatality rate	δ_θ	**0.009** (0.9%)	0.008–0.010 (pre-vax)	Levin et al. (2020); Meyerowitz-Katz & Merone (2020)
GDP impact of lockdown	α_S	**12** (pp)	8–15	Deb et al. (2021); Coibion et al. (2020); Born et al. (2021)
NPI effect on R_t	φ_S	**0.70**	0.60–0.82	Brauner et al. (2021); Flaxman et al. (2020)
Transfer multiplier	α_F_DI	**0.50** (impact)	0.30–1.00	Parker et al. (2022); Coibion et al. (2020); CBO (2020)

Parameter	Symbol	Central value	Range	Key sources
DI budgetary cost	κ_F_DI	**0.80**	0.70–0.90	CBO analysis; OECD Revenue Statistics
CP effectiveness	η	**0.05**	0.03–0.08	Giupponi & Landais (2023); Autor et al. (2022)
CP budgetary cost	κ_F_CP	**0.60**	0.50–0.70	SBA PPP data; UK OBR; France PGE data
Real interest rate (quarterly)	r	**−0.25%**	−0.38% to −0.13%	OECD bond yields; Rachel & Summers (2019)
Budget semi-elasticity	γ_y	**0.50**	0.42–0.55	Price et al. (2015); Mourre et al. (2019)
Health expenditure rate	c_H	**10**	5–20	OECD Health at a Glance 2021; WHO health spending data

Conclusion: what the calibration reveals about pandemic policy trade-offs

Three features of this calibration stand out. First, the extreme asymmetry between ρ_θ (infection growth) and φ_S (NPI effectiveness) means that even modest relaxation of containment produces explosive infection dynamics — the epidemiological constraint binds tightly. Second, the pandemic fiscal multiplier for transfers (**~0.5**) was strikingly low compared to normal recessions, while capacity preservation was relatively cost-effective (κ_F_CP ≈ 0.60 vs. κ_F_DI ≈ 0.80) and directly targeted the scarring channel. This supports the European policy model of prioritizing worker-firm match preservation over aggregate demand stimulation during a supply-constrained crisis. Third, deeply negative real interest rates (r ≈ −0.25% quarterly) dramatically lower the true cost of debt-financed pandemic spending, making aggressive fiscal intervention nearly self-financing through the r < g channel documented by Blanchard (2019).

The parameters where **direct estimation is weakest** are ψ (lockdown hysteresis), η (capacity preservation effectiveness), ρ_θ (quarterly infection growth in the reduced-form model), and c_H (health expenditure per prevalence). For these, moment-matching calibration — choosing values so the model replicates observed trajectories of GDP, infections, mortality, and debt across OECD countries in 2020–2021 — is the recommended strategy. The remaining parameters are well-identified in the empirical literature and provide a solid foundation for the optimal control analysis.